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### Image forming apparatus

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#### Abstract

According to one embodiment, an image forming apparatus has a printing unit to print images on a sheet of paper from a paper storage unit and a conveying unit configured to convey sheets from the storage unit to the printing unit. A tag reader/writer communicates with a wireless tag on a sheet when the sheet is at a communication position on the conveyance path. A control unit determines whether a writing of data to a tag on a first sheet has succeeded, then controls the printing unit to print on the first sheet if the writing of data succeeded but not if the writing of data to the wireless tag failed. When writing of data to the wireless tag has failed, the control unit executes a tag data writing retry process on another sheet from the paper storage unit.

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**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-102646, filed Jun. 22, 2023, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to an image forming apparatus.

BACKGROUND

(3) An image forming apparatus having an RFID reader/writer function is capable of not only forming an image on paper in which an RFID tag is embedded but also writing data to the RFID tag.

(4) In the related art, when a tag data writing error occurs with an RFID-tagged paper, ink may be

saved by not printing an image on the paper. However, the tagged paper for which the data writing error occurs is usually not reused.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional of an image forming apparatus according to a first embodiment.
- (2) FIG. 2 is a perspective view of a paper discharge unit of an image forming apparatus.
- (3) FIG. 3 is a block diagram of a control unit of an image forming apparatus.
- (4) FIG. 4 is a block diagram of a processor.
- (5) FIG. 5 is a flowchart illustrating an example of RFID processing by an image forming apparatus.
- (6) FIG. 6 is a flowchart illustrating an example of RFID writing by an image forming apparatus according to a second embodiment.
- (7) FIG. 7 is a flowchart illustrating an example of conveyance control processing in RFID processing of an image forming apparatus.

### DETAILED DESCRIPTION

- (8) An image forming apparatus of one embodiment notifies a user of the paper on which an RFID data writing error occurs to permit the paper to be reused.
- (9) According to one embodiment, an image forming apparatus includes a paper storage unit, a printing unit configured to print images on a sheet of paper from the paper storage unit, a conveying unit configured to convey sheets one by one along a path from the paper storage unit to the printing unit, a tag reader/writer unit configured to communicate with a wireless tag on a sheet being conveyed by the conveying unit when the sheet is at a communication position on the path before the printing unit, and a control unit. The control unit is configured to determine whether a writing of first data to a wireless tag on a first sheet has succeeded by controlling the tag reader/writer to read data from the wireless tag on the first sheet, control the printing unit to print an image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet succeeded, control the printing unit not to print any image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet failed, and, after it is determined the writing of the first data to the wireless tag on the first sheet failed, execute a retry process by controlling the conveying unit to convey a second sheet from paper storage unit to the communication position on the path and the tag reader/writer to write the first data to a wireless tag on the second sheet.
- (10) Hereinafter, an image forming apparatus according to certain example embodiments will be described with reference to the drawings. It is noted that in the drawings, the depicted dimensions, relative scale, and the like of each component may be varied as appropriate for purposes of explanation. Further, in each drawing, aspects and components may be omitted as appropriate for purposes explanatory convenience.

#### First Embodiment

- (11) FIG. 1 is a schematic cross-sectional view illustrating an example of an image forming apparatus **100** according to a first embodiment. The image forming apparatus **100** is a digital multifunctional peripheral (MFP) and includes a scanner **1**, a printer **2**, an operation panel **4**, an RFID reader/writer **5**, and a control unit **6**.
- (12) The scanner **1** is a device that reads (scans) an image from a document and converts the scanned image into image data. The scanner **1** includes, for example, a charge coupled device (CCD) line sensor that converts the image on the document into the image data. The scanner **1** may have the function of scanning the document placed on a document table (platen) glass. Further, the

scanner **1** may have a function of reading the image of the document as it is conveyed by an auto document feeder (ADF). The scanner **1** is installed, for example, on the top of a main body of the MFP. The scanner **1** is controlled by the control unit **6**. The scanner **1** outputs the image data of the document to control unit **6**.

(13) The printer **2** is an electrophotographic printer in this example. The printer **2** forms (prints) the image on paper or other recording medium. The printer **2** has a color printing function that prints a color image on the paper and a monochrome printing function that prints a monochrome (for example, black) image on the paper. The printer **2** forms the color image by using toner of a plurality of colors (for example, three colors of yellow (Y), cyan (C), and magenta (M)). Further, the printer **2** forms a monochrome image by using monochrome (for example, black (K)) toner.

(14) In the configuration example illustrated in FIG. **1**, the printer **2** includes paper feed cassettes **20A**, **20B**, and **20C**. The paper feed cassettes **20A**, **20B**, and **20C** that supply paper. Each of these paper feed cassettes **20A**, **20B**, and **20C** stores paper of a particular type (for example, size, paper quality). Furthermore, the printer **2** also has a manual feed tray **20D**. The manual feed tray **20D** may receive/supply any type of paper (for example, any size or paper quality). The paper feed cassettes **20A**, **20B**, **20C** and the manual feed tray **20D** may be referred to individual or collective as a paper feed unit. The paper supplied by each paper feed cassette **20A**, **20B**, **20C** and the manual feed tray can be regular paper P.sub.st of any size or RFID-tagged paper P.sub.rf (also referred to as RFID paper P.sub.rf). An RFID tag is embedded in the RFID paper P.sub.rf. In the example illustrated in FIG. **1**, the RFID paper P.sub.rf is set in the paper feed cassette **20A**.

(15) The paper feed cassettes **20A**, **20B**, and **20C** have pick-up rollers **21A**, **21B**, and **21C**, respectively. The pick-up rollers **21A**, **21B**, and **21C** pick up the paper one by one from the respective paper feed cassettes **20A**, **20B**, and **20C**. Similarly, the manual feed tray **20D** is also provided with a pick-up roller **21D** that feeds the paper one by one from the manual feed tray **20D**. These pick-up rollers **21A**, **21B**, **21C**, and **21D** supply the paper to the conveyance path (conveying unit **22**) configured with a plurality of conveyance rollers **22A**, **22B**, **22C**, **22D**, and the like.

(16) The conveying unit **22** conveys the paper within the printer **2**. For example, the conveying unit **22** conveys the paper picked up by one of the pick-up rollers **21A**, **21B**, **21C**, and **21D** to a registration roller **23**. The registration roller **23** corrects an inclination (angle) of the paper so the paper is substantially perpendicular to the conveyance direction by stopping the conveyed paper against the registration roller **23**. The registration roller **23** then conveys the paper to the transfer position at the appropriate timing for transferring the image from a transfer belt **26** to the paper. The conveying unit **22** conveys the paper through the registration roller **23** to the transfer position. The conveying unit **22** conveys the paper from the transfer position to a fixer **28**. The conveying unit **22** conveys the paper after the fixer **28** to either a paper discharge unit **29** or an automatic double-sided unit (ADU). The conveying unit **22** includes a flapper **22E** in front of the paper discharge unit **29**. The flapper **22E** at a position between the first paper discharge unit and the second paper discharge unit of the paper discharge unit **29** switches a discharge destination.

(17) FIG. **2** is a perspective view illustrating an example of the paper discharge unit **29**. As illustrated in FIG. **2**, the paper discharge unit **29** is provided with a lower stage paper discharge tray **29A** that functions as a first paper discharge unit and an upper stage paper discharge tray **29B** that functions as a second paper discharge unit.

(18) The printer **2** has a plurality of image forming units **24Y**, **24M**, **24C**, and **24K**. These plurality of image forming units **24Y**, **24M**, **24C**, and **24K** form images to be transferred to the paper. In the configuration example illustrated in FIG. **1**, the image forming unit **24Y** forms the image with yellow toner. The image forming unit **24M** forms the image with magenta toner. The image forming unit **24C** forms the image with cyan toner. The image forming unit **24K** forms the image with black toner.

(19) The image forming units **24Y**, **24M**, **24C**, and **24K** include photosensitive drums **30y**, **30m**, **30c**, and **30k**, chargers **31y**, **31m**, **31c**, and **31k**, developing devices **32y**, **32m**, **32c**, and **32k**, and

transfer rollers **33y**, **33m**, and **33c**, **33k**, and cleaners **34y**, **34m**, **34c**, and **34k**. Since, in general, the only difference is color, unless there is the particular need to distinguish between image forming units **24Y**, **24M**, **24C**, and **24K**, each may be referred to as an image forming unit **24** and similarly for each photosensitive drum **30**, charger **31**, developing device **32**, transfer roller **33**, and cleaner **34**.

(20) The photosensitive drum **30** is an image carrier on which the electrostatic image latent is formed. The photosensitive drum **30** is rotated by a rotating shaft. The charger **31** charges the surface of the photosensitive drum **30** to a predetermined potential. The charger **31** has a grid for adjusting the charging output to the photosensitive drum **30**. The developing device **32** develops the electrostatic latent image formed on the photosensitive drum **30** with the toner. The transfer roller **33** transfers a toner image developed on the photosensitive drum **30** onto the transfer belt **26**. The cleaner **34** cleans the surface of the photosensitive drum **30** after the transfer.

(21) An exposure unit **25** forms the electrostatic latent images on the photosensitive drums **30** of each of the image forming units **24Y**, **24M**, **24C**, and **24K** by using a laser beam. The exposure unit **25** irradiates the photosensitive drum **30** with the laser beam that is controlled according to the image data using the optical system such as a polygon mirror. The laser beam from the exposure unit **25** forms the electrostatic latent image on the surface of each photosensitive drum **30**. The exposure unit **25** controls the laser beam according to a control signal from the control unit **6**.

(22) The image forming unit **24** develops the electrostatic latent image formed on the photosensitive drum **30** by the developing device **32**. The developing device **32** includes a developer container having a developing roller. The developer container stores toner of the particular color of the image forming unit **24**. The toner is stirred together with a carrier in the developer container. A developing bias is applied to the developing roller. The biased developing supplies the toner to the electrostatic latent image on the photosensitive drum **30**. The electrostatic latent image on the photosensitive drum **30** becomes the toner image transferred to the paper or the like.

(23) The transfer belt **26** is an intermediate transfer body. The image forming unit **24** transfers (primary transfer) the toner image from the photosensitive drum **30** onto the transfer belt **26** by applying the primary transfer voltage to the transfer belt **26** by using the transfer roller **33**. For example, in the image forming unit **24K**, the transfer roller **33k** transfers the black toner image formed by the developing device **32k** to the transfer belt **26**. Further, when forming the color image, each of the image forming units **24Y**, **24M**, **24C**, and **24K** transfers the toner images of each color to the transfer belt **26** in an overlapping manner.

(24) The transfer unit **27** transfers the toner image from the transfer belt **26** to the paper at the secondary transfer position. In other words, the transfer unit **27** forms the toner image on the paper. The transfer unit **27** can be referred as a printing unit. The secondary transfer position, which may be referred to as a printing position, is where the toner image is transferred to the paper from the transfer belt **26**. The secondary transfer position is where a support roller **27a** and a secondary transfer roller **27b** face each other.

(25) The fixer **28** fixes the toner (toner image) to the paper. The fixer **28** applies heat to the paper. In the example illustrated in FIG. 2, the fixer **28** has a heat roller **28b** (having a built-in heating unit **28a**) and a pressure roller **28c** (pressing roller) that is in contact with a fixing belt that is heated by the heat roller **28b**. The heating unit **28a** may be any heater by which temperature can be controlled. For example, the heating unit **28a** may be a heater lamp such as a halogen lamp or may be an induction heating (IH) type heater. Further, the heating unit **28a** may comprise a plurality of heaters. The fixer **28** conveys the paper after the fixing processing to one of the paper discharge unit **29** or the ADU.

(26) The operation panel **4** is a user interface (operator panel). The operation panel **4** includes various buttons and a display unit **4a** including a touch panel **4b**. The control unit **6** controls contents displayed on the display unit **4a** of the operation panel **4**. The display unit **4a** functions as

a notification unit. Further, the operation panel **4** provides information to the control unit **6** as input to the touch panel **4b** or the buttons of the operation panel **4**. The user designates the operation mode of the image forming apparatus **1** and inputs other information such as setting information (e.g., print parameters or the like) on the operation panel **4**.

(27) The RFID reader/writer **5** includes a controller (also referred as an RFID module **51**) and an antenna **52**. The RFID module **51** writes the data from the control unit **6** via the antenna **52** to the RFID tag embedded in a piece of the RFID paper P.sub.rf while the paper is stopped by the registration roller **23**. Further, the RFID module **51** reads data already written in the RFID tag via the antenna **52** and outputs the read data to the control unit **6**. The positioning and transmitting/receiving power of the RFID reader/writer **5** are set so that the transmitting/receiving range of the antenna **52** covers the RFID tag of the RFID paper P.sub.rf when stopped at the registration roller **23**.

(28) An example of a configuration of the control unit **6** of the image forming apparatus **100** will be described with reference to FIG. **3**. FIG. **3** is a block diagram illustrating the control unit **6**. The control unit **6** includes, for example, a processor **61**, a read-only memory (ROM) **62**, a random-access memory (RAM) **63**, an auxiliary storage device **64**, a scanner interface **65**, a printer interface **66**, an input/output interface **67**, RFID reader/writer interface **68**, a communication interface **69**, and a real-time clock (RTC) **70**.

(29) The processor **61** performs processing such as computation and control necessary for the operation of the image forming apparatus **100**. The processor **61** controls components to realize the various functions of the image forming apparatus **100** based on programs such as system software, application software, or firmware stored in the ROM **62** or the auxiliary storage device **64**. The processor **61** is, for example, a central processing unit (CPU), a micro processing unit (MPU), a system on a chip (SoC), a digital signal processor (DSP), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), a programmable logic device (PLD), a field-programmable gate array (FPGA), or the like. In some examples, the processor **61** may be a combination of a plurality of these devices.

(30) The ROM **62** is a non-volatile computer-readable storage medium. The ROM **62** stores data or various setting values used by the processor **61** to perform various processes.

(31) The RAM **63** is volatile memory. The RAM **63** is used as a so-called work area or the like for temporarily storing data when the processor **61** performs various processes.

(32) The auxiliary storage device **64** is a non-volatile computer-readable storage medium. The auxiliary storage device **64** is, for example, an electric erasable programmable read-only memory (EEPROM®), a hard disk drive (HDD), or a solid-state drive (SSD). The auxiliary storage device **64** stores data used by the processor **61** to perform various processes, data generated with processing by the processor **61**, various setting values, and the like.

(33) It is noted that the image forming apparatus **100** may be provided with the interface into which the storage medium such as a removable optical disk, a memory card, or a universal serial bus (USB) memory can be used instead of the auxiliary storage device **64** or in addition to the auxiliary storage device **64**.

(34) The programs stored in the ROM **62** or the auxiliary storage device **64** may be the programs with instructions for executing the processing described below. As an example, the image forming apparatus **100** is transferred with the program stored in the ROM **62** or the auxiliary storage device **64**. In other examples, the image forming apparatus **100** may be transferred without the program being pre-stored in the ROM **62** or the auxiliary storage device **64** and such a program may be separately transferred or the like and then written into the ROM **62** or the auxiliary storage device **64**. The program may be transferred, for example, by recording the program on a removable storage medium such as a magnetic disk, a magneto-optical disk, an optical disk, or a semiconductor memory, or by downloading the program via a network.

(35) The scanner interface **65** is an interface for the scanner **1**. The scanner interface **65** transmits

control signals output from the processor **61** to the scanner **1**. The scanner interface **65** receives the image data of a document and operational state signals from the scanner **1**, and provides the operational state signal(s) to the processor **61**.

(36) The printer interface **66** is an interface for the printer **2**. The printer interface **66** transmits control signals output from the processor **61** to the printer **2**. Further, the printer interface **66** receives operating state signals from the printer **2** and provides the operating state signal(s) to the processor **61**.

(37) The input/output interface **67** is an interface for the operation panel **4**. The input/output interface **67** transmits information to be displayed on the display unit **4a** to the operation panel **4** from the processor **61**. The input/output interface **67** also receives information input via the touch panel **4b** or buttons on the operation panel **4** and provides this information to the processor **61**.

(38) The RFID reader/writer interface **68** is an interface for the RFID reader/writer **5**. The RFID reader/writer interface **68** transmits data to be written to the RFID tag of RFID paper P.sub.rf to the RFID reader/writer **5**. Furthermore, the RFID reader/writer interface **68** receives data that has been read from the RFID tag by the RFID reader/writer **5** to the processor **61**.

(39) The communication interface **69** can be a wired or wireless interface for communicating with a server device and/or a user terminal via a network, or the like. The communication interface **69** is used for receiving data, such as image data to be printed and the data to be written to the RFID tag, and control information such as that related to the selection of printing paper and the number of copies to be printed or the like.

(40) The RTC **70** is a clock or a circuit with a built-in clock function.

(41) FIG. **4** is a block diagram illustrating an example of the functional configuration aspects of the processor **61**. The processor **61** provides an RFID paper conveyance control unit **611**, an RFID writing control unit **612**, a writing success/failure determination unit **613**, and an RFID paper printing control unit **614**. For example, the processor **61** executes a program stored in the ROM **62** or the auxiliary storage device **64** to realize the described functions of the RFID paper conveyance control unit **611**, RFID writing unit control **612**, writing success/failure determination unit **613**, and RFID paper printing control unit **614**. It is noted that the RFID paper conveyance control unit **611**, the RFID writing control unit **612**, the writing success/failure determination unit **613**, and the RFID paper printing control unit **614** may also or instead be realized by hardware such as large scale integration (LSI), an application specific integrated circuit (ASIC), and a field-programmable gate array (FPGA) providing the same functions as those described for the processor **61** executing a software program or the like.

(42) The RFID paper conveyance control unit **611** controls the conveying of the RFID paper P.sub.rf by controlling the driving of each roller of the conveying unit **22** of the printer **2** via the printer interface **66**.

(43) The RFID writing control unit **612** controls the RFID reader/writer **5** via the RFID reader/writer interface **68** to control writing of data to the RFID tag of the RFID paper P.sub.rf.

(44) The writing success/failure determination unit **613** determines whether the writing has succeeded or failed by comparing the read data input from the RFID reader/writer **5** via the RFID reader/writer interface **68** with the data that was instructed to be written. It is noted that, in some examples, the RFID reader/writer **5** has a write data verification function, and the writing success/failure determination unit **613** may not need to match read/write data, but may rather identify the success or failure of a writing based on a verification result provided from the RFID reader/writer **5** via the RFID reader/writer interface **68**.

(45) The RFID paper printing control unit **614** controls printing, that is, image formation on the RFID paper P.sub.rf depending on whether the writing of data to the RFID tag succeeded.

Specifically, the RFID paper printing control unit **614** performs printing on the RFID paper P.sub.rf only after the writing succeeds and controls not to print if the writing fails.

(46) It is noted that, based on the printing control by the RFID paper printing control unit **614**, the

RFID paper conveyance control unit **611** controls the discharge destination of the RFID paper P.sub.rf to switch between the lower stage paper discharge tray **29A** and the upper stage paper discharge tray **29B**.

(47) Hereinafter, the operation of the image forming apparatus **100** according to the first embodiment will be described with reference to FIG. 5. It is noted that the specifics of the processing in the following description are merely examples, and various processing modifications that can obtain similar effects can be used as appropriate.

(48) FIG. 5 is a flowchart illustrating the example of RFID processing related to the writing data to, and the printing on, the RFID paper P.sub.rf according to the first embodiment. The processor **61** performs this processing based on the program stored in the ROM **62**, auxiliary storage device **64**, or the like.

(49) When the image forming apparatus **100** receives the image data to be printed on the RFID paper P.sub.rf and the data to be written to the RFID tag from a server device or a user terminal via the communication interface **69**, the image forming apparatus **100** starts the process illustrated in FIG. 5. It is noted that the image data and data to be written can be stored in the RAM **63** or the auxiliary storage device **64**.

(50) In ACT11, the processor **61** operates as the RFID paper conveyance control unit **611** and controls the printer **2** via the printer interface **66** to convey the RFID paper P.sub.rf. For example, in the example illustrated in FIG. 1, the processor **61** causes the pick-up roller **21A** to pick up a piece of RFID paper P.sub.rf from the paper feed cassette **20A** and convey the RFID paper P.sub.rf to the registration roller **23**.

(51) In ACT12, the processor **61** determines whether the RFID paper P.sub.rf reaches a writing position. The writing position corresponds to the position of the registration roller **23** in this first embodiment. Therefore, in this example, it is determined whether the RFID paper P.sub.rf reaches the registration roller **23**. For example, the image forming apparatus **100** can have a detector, such as a mechanical switch or an optical sensor, for detecting when paper reaches or is at the registration roller **23**. The processor **61** can acquire the output of such a detector via the printer interface **66** or the like. If the RFID paper P.sub.rf does not reach the writing position (NO in ACT12), the processor **61** repeats the processing operation in ACT12. In this manner, the processor **61** waits for the RFID paper P.sub.rf to reach the registration roller **23**.

(52) When the RFID paper P.sub.rf reaches the writing position (YES in ACT12), the processor **61** controls the conveying unit **22** via the printer interface **66** in ACT13 to temporarily stop the conveying of the RFID paper P.sub.rf.

(53) In ACT14, the processor **61** operates as the RFID writing control unit **612** and writes RFID data. Specifically, the processor **61** transmits the data to be written to the RFID tag of the RFID paper P.sub.rf as stored in the RAM **63** or the auxiliary storage device **64** to the RFID module **51** of the RFID reader/writer **5** via the RFID reader/writer interface **68**. The RFID module **51** upon receiving this data executes writing to the RFID tag of the RFID paper P.sub.rf that is stopped at the writing position. Specifically, the RFID module **51** emits electromagnetic waves for writing data (write information) by using the antenna **52**. By receiving this electromagnetic wave, the RFID tag of the RFID paper P.sub.rf writes data to the storage area.

(54) In ACT15, the processor **61** operates as the writing success/failure determination unit **613** and acquires (as read data) the data that was written to the RFID tag of the RFID paper P.sub.rf. Specifically, the processor **61** instructs the RFID reader/writer **5** to read data from the RFID tag via the RFID reader/writer interface **68**. Upon receiving this instruction, the RFID module **51** emits electromagnetic waves from the antenna **52** for reading data written in the RFID tag. Upon receiving this electromagnetic wave, the RFID tag responds with the data recorded in the storage area of the RFID tag. The RFID module **51** outputs data (the read data) received from the RFID tag via the antenna **52**. The processor **61** acquires the read data from the RFID module **51** via the RFID reader/writer interface **68**.



(55) In ACT16, the processor **61** determines whether the writing of data to the RFID tag of the RFID paper P.sub.rf has succeeded. Specifically, whether the writing succeeds or fails is determined by comparing the data as read (the read data) from the RFID reader/writer **5** via the RFID reader/writer interface **68** matches the data to be written that was stored in the RAM **63** or the auxiliary storage device **64**. If the writing does not succeed, that is, the writing fails (NO in ACT16), the processor **61** shifts to the processing operation of ACT22

(56) If the writing succeeds (YES in ACT16), the processor **61** operates as the RFID paper conveyance control unit **611** in ACT17 and sets the first conveyance path as the conveyance path for the successfully written RFID paper P.sub.rf. Specifically, the processor **61** switches the flapper **22E** in the conveying unit **22** to the lower stage paper discharge tray **29A** via the printer interface **66**.

(57) In ACT18, the processor **61** operates as the RFID paper printing control unit **614** and forms (prints) the image. Specifically, the processor **61**, via the printer interface **66**, causes the image forming unit **24** to transfer (primarily transfer) the toner image generated according to the image data to the transfer belt **26**.

(58) In ACT19, the processor **61** resumes the conveying of the RFID paper P.sub.rf, which was temporarily stopped at the position of the registration roller **23**. Specifically, the processor **61** restarts driving of rollers of the conveying unit **22** via the printer interface **66**. Accordingly, the temporarily stopped RFID paper P.sub.rf is conveyed to the transfer position at the timing appropriate for transferring the image from the transfer belt **26** to the RFID paper P.sub.rf, and thus, the toner image transferred on the transfer belt **26** is secondarily transferred onto the RFID paper P.sub.rf. In this manner, the image is printed on the RFID paper P.sub.rf after data was successfully written to the RFID tag.

(59) In ACT20, the processor **61** determines whether the RFID paper P.sub.rf for which the RFID data was successfully and on which an image has been printed is conveyed through the first conveyance path and discharged to the lower stage paper discharge tray **29A**. For example, the image forming apparatus **100** can include a detector such as a mechanical switch or an optical sensor for detecting that the paper is discharged. The processor **61** can acquire the output of such a detector/sensor via the printer interface **66**. If the RFID paper P.sub.rf is not yet discharged to the lower stage paper discharge tray **29A** (NO in ACT20), the processor **61** repeats the processing operation of this ACT20. In this manner, the processor **61** waits for the RFID paper P.sub.rf to be discharged to the lower stage paper discharge tray **29A**.

(60) Once the RFID paper P.sub.rf is discharged to the lower stage paper discharge tray **29A** (YES in ACT20), the processor **61** determines in ACT21 whether the processing is to end. Specifically, the processor **61** can perform this termination determination by checking whether other data is to be written to the RFID paper P.sub.rf is stored in the RAM **63** or the auxiliary storage device **64**. Alternatively, the processor **61** can check whether the writing of data to the paper has been completed. If the processing is not finished (NO in ACT21), the processor **61** returns to the processing operation of ACT11. Accordingly, the above-described processing operation can be repeated for the next RFID paper P.sub.rf. In this manner, the RFID paper P.sub.rf on which the data is written to the RFID tag and the image is printed is placed on the lower stage paper discharge tray **29A**.

(61) If the process is complete in ACT21 (YES in ACT21), the processor **61** ends the processing illustrated in this flowchart.

(62) If the writing fails in ACT16 (NO in ACT16), the processor **61** operates as the RFID paper conveyance control unit **611** in ACT22 and sets the second conveyance path as the conveyance path for the RFID paper P.sub.rf (an unsuccessfully written RFID paper P.sub.rf). Specifically, the processor **61** switches the flapper **22E** in the conveying unit **22** to the upper stage paper discharge tray **29B** side via the printer interface **66**.

(63) In ACT23, the processor **61** resumes the conveying of the RFID paper P.sub.rf which was

temporarily stopped at the position of the registration roller **23**. Specifically, the processor **61** restarts driving of rollers of the conveying unit **22** of the printer **2** via the printer interface **66**. Accordingly, the temporarily stopped RFID paper P.sub.rf is conveyed toward the upper stage paper discharge tray **29B**. At this time, the processor **61** operates as the RFID paper printing control unit **614** and does not permit the image forming unit **24** to perform printing operations on the paper. Therefore, even when the RFID paper P.sub.rf passes through the secondary transfer position, no printing is performed on the RFID paper P.sub.rf and the RFID paper P.sub.rf remains blank (unprinted).

(64) In ACT**24**, the processor **61** determines whether the blank RFID paper P.sub.rf is conveyed through the second conveyance path and discharged to the upper stage paper discharge tray **29B**. If the blank RFID paper P.sub.rf is not yet discharged to the upper stage paper discharge tray **29B** (NO in ACT**24**), the processor **61** repeats the processing operation of this ACT**24**. In this manner, the processor **61** waits for the blank RFID paper P.sub.rf to be discharged to the upper stage paper discharge tray **29B**.

(65) Once the blank RFID paper P.sub.rf is discharged to the upper stage paper discharge tray **29B** (YES in ACT**24**), the processor **61** operates as the writing success/failure determination unit **613** in ACT**25** and notifies of a writing error. Specifically, the processor **61** performs an error display on the display unit **4a** of the operation panel **4** via the input/output interface **67**. The error display notification may indicate a blank RFID paper P.sub.rf has been discharged to the upper stage discharge tray **29B** or the like. When the image forming apparatus **100** includes a speaker as an output device, an error sound may be output together with this error display. Furthermore, or alternatively, the processor **61** can transmit an error notification via the communication interface **69** to a server device or user terminal which was a transmission source of the data to be written.

(66) After that, the processor **61** returns to the processing operation of ACT**11**. Accordingly, since the above-described processing can be repeated for the next RFID paper P.sub.rf, the writing of data that was not successfully written can be attempted on the next RFID paper P.sub.rf. In this manner, the blank RFID paper P.sub.rf on which the RFID data could not be written is placed on the upper stage paper discharge tray **29B**, and a retry process of writing the RFID data to another RFID paper P.sub.rf can be executed.

(67) As described above, the image forming apparatus **100** according to the first embodiment includes a conveying unit **22** that conveys the RFID paper P.sub.rf from the paper feed cassette (for example, the paper feed cassette **20A**) or the manual feed tray **20D**. The RFID paper P.sub.rf has a RFID tag and may be of any physical size. A transfer unit **27** transfers the toner image from the transfer belt **26** to the RFID paper P.sub.rf conveyed by the conveying unit. A RFID reader/writer **5** communicates with the RFID tag embedded in the RFID paper P.sub.rf to write data to the RFID tag and also read data written in the RFID tag. The RFID paper P.sub.rf is stopped at a position between the paper feed cassette (or the manual feed tray **20D**) and the transfer unit **27** when the RFID tag is written or read. A processor **61** of the control unit **6** that controls the conveying unit **22**, image forming unit **24**, transfer belt **26**, and transfer unit **27**, and the RFID reader/writer **5**. In this manner, a paper feed cassette (for example, the paper feed cassette **20A**) and a manual feed tray **20D** are paper placing units, the image forming unit **24**, the transfer belt **26**, and the transfer unit **27** are printing units, and the RFID reader/writer **5** is a reader/writer unit. According to an image forming apparatus **100** according to the first embodiment, the processor **61** determines whether the writing data to the RFID tag has succeeded based on the data read by the RFID reader/writer **5**. When it is determined that the writing data has succeeded, the processor **61** controls the printing unit so as to perform printing on the RFID paper P.sub.rf. If the processor **61** determines that writing of data fails, the processor **61** controls the printing unit so as not to print on the RFID paper P.sub.rf and controls the conveying unit **22** and the RFID reader/writer **5** to convey another (different) RFID paper P.sub.rf from the paper placing unit and executes a retry process for permitting the previously failed tag data to be written to the RFID tag in another (next) RFID paper

P.sub.rf to be fed.

(68) There are cases where the RFID tag may have been damaged by impact or the like and become unusable, or the tag damage may occur in the printing on the RFID paper P.sub.rf rather than the writing of data to RFID tag. Therefore, in an image forming apparatus **100** continuously prints on a roll of paper before individual sheets are cut from the roll or the like, when the writing error occurs before printing, the RFID paper P.sub.rf (cut sheet) can be discharged as a blank sheet without being printed. In this manner, by not performing printing, that is, by not performing unnecessary printing, the toner can be saved.

(69) Furthermore, by not performing printing on a RFID paper P.sub.rf for which the RFID tag writing failed, the next printed RFID paper P.sub.rf (the replacement sheet) for which RFID tag data was correctly written can be easily distinguished from the blank (unwritten RFID paper P.sub.rf). The user of the image forming apparatus **100** can recognize whether there is the RFID tag writing error by checking whether the paper is blank, and since the paper is blank, any RFID paper P.sub.rf for which a RFID tag writing error occurs can be reused as regular paper (a sheet without the RFID). In this manner, an image forming apparatus that notifies the user of a failed RFID paper P.sub.rf allows the failed RFID paper P.sub.rf to be reused.

(70) It is noted that there are various factors that cause the tag writing errors, and there may be cases where the RFID tag in the RFID paper P.sub.rf does not itself fail. For example, there may be an error caused by external radio waves (temporary interference), a failure or glitch of the RFID reader/writer **5** on the image forming apparatus **100** side, or the incorrect use/loading of regular paper (paper without the RFID tag). In such cases, the RFID paper P.sub.rf will still be blank after a failed tag writing attempt. Therefore, the RFID paper P.sub.rf discharged as blank paper may be reused not only as regular paper (paper without RFID) but as RFID paper P.sub.rf again. When the RFID tag of the RFID paper P.sub.rf is out of order, the paper is simply discharged as the blank paper again.

(71) Furthermore, when the tag writing error occurs, since the processor **61** retries writing the data and printing the image for which the writing data fails on another RFID paper P.sub.rf, the data requested to be written and printed and the RFID paper P.sub.rf with the image recorded thereon can still be reliably obtained.

(72) It is noted that the image forming apparatus **100** according to the first embodiment is provided with the lower stage paper discharge tray **29A** and the upper stage paper discharge tray **29B**, and the conveying unit **22** is provided with the flapper **22E** for switching the discharge destination of the RFID paper P.sub.rf between the lower stage paper discharge tray **29A** and the upper stage paper discharge tray **29B**. In this manner, the lower stage paper discharge tray **29A** is a first paper discharge unit. The upper stage paper discharge tray **29B** is a second paper discharge unit, and the flapper **22E** is a switching unit. The processor **61** of the control unit **6** controls the flapper **22E** so that the RFID paper P.sub.rf on which an image has been printed is discharged to the lower stage paper discharge tray **29A** and the RFID paper P.sub.rf on which the image has not been printed is discharged to the upper stage paper discharge tray **29B**.

(73) In this manner, a user can distinguish between the RFID paper P.sub.rf with a tag writing error and the RFID paper P.sub.rf on which the tag writing was normally performed by the paper discharge tray branching.

(74) The image forming apparatus **100** according to the first embodiment is further provided with a display unit **4a** (of the operation panel **4**) and/or a communication interface **69** that also notifies the user when writing data to a RFID tag fails and failed RFID paper P.sub.rf is discharged to the upper stage paper discharge tray **29B**. In this manner, the display unit **4a** or the communication interface **69** is a notification unit.

(75) The image forming apparatus **100** according to the first embodiment includes the registration roller **23** that is disposed in front of the transfer unit **27** of the printing unit and temporarily stops the RFID paper P.sub.rf conveyed from the paper placing unit, and the RFID reader/writer **5** is

placed at a position near where the RFID paper P.sub.rf is temporarily stopped by the registration roller **23** for communicating with the RFID tag in the temporarily stopped RFID paper P.sub.rf. Then, the processor **61** of the control unit **6** controls the RFID reader/writer **5** so as to perform the writing data to the RFID tag and the reading data from the RFID tag when the RFID paper P.sub.rf is temporarily stopped by the registration roller **23**.

(76) In this manner, since writing and reading of data to and from the RFID tag by the RFID reader/writer **5** can be performed with the RFID paper P.sub.rf being stopped, reliable writing data and reading can be realized.

(77) Further, in the image forming apparatus **100** according to the first embodiment, the processor **61** of the control unit **6** controls the conveying unit **22** so that the conveying of the RFID paper P.sub.rf is resumed after waiting for the determination of the success or failure of the writing of data to the RFID tag.

(78) In this manner, the RFID paper P.sub.rf is kept at the position of the registration roller **23** until the determination of the success or failure of writing data to the RFID tag is ascertained, so that there is no concern that an image will be printed on RFID paper P.sub.rf for which writing tag data is not successfully completed.

#### Second Embodiment

(79) In the first embodiment, the processing operation for the next RFID paper Prf is started after the prior RFID paper Prf is discharged. This second embodiment is an example in which this waiting is omitted.

(80) The image forming apparatus **100** according to the second embodiment is generally the same as that of the first embodiment, and overlapping description will be omitted.

(81) FIG. **6** is a flowchart illustrating an example of RFID writing by the image forming apparatus **100** according to the second embodiment, and FIG. **7** is a flowchart illustrating an example of conveyance control processing in the RFID processing. The processor **61** can execute these two processes simultaneously in the second embodiment.

(82) In the RFID writing processing operation illustrated in FIG. **6**, processing in ACT**11** to ACT**16** is the same as in the first embodiment.

(83) When the writing is determined in ACT**16** (YES in ACT**16**) to have succeeded, the processor **61** operates in ACT**18** as the RFID paper printing control unit **614** to form an image. In ACT**19**, the processor **61** resumes the conveying of the RFID paper P.sub.rf, which was temporarily stopped at the registration roller **23** (tag writing position). Accordingly, the image is printed on the RFID paper P.sub.rf after data is written to the RFID tag.

(84) In ACT**21**, the processor **61** determines whether to end the present processing.

(85) If not (NO in ACT **21**), the processor **61** operates as the RFID paper conveyance control unit **611** in ACT **31** and controls the printer **2** via the printer interface **66** to start the conveying of the next RFID paper P.sub.rf. For example, in the example illustrated in FIG. **1**, the processor **61** causes the pick-up roller **21A** to pick up the next piece of paper of the RFID paper P.sub.rf from the paper feed cassette **20A** and convey the just fed RFID paper P.sub.rf to the registration roller **23**. After that, the processor **61** returns to the processing operation of ACT**12**. Accordingly, at the same time as a printing on an RFID paper P.sub.rf for which an RFID tag has been successfully written with data and the conveying of this printed sheet to the lower stage paper discharge tray **29A**, initial processing operations for the next RFID paper P.sub.rf to be printed can be performed.

(86) Based on a determination that the writing fails in ACT**16** (NO in ACT**16**), the processor **61** operates as the RFID paper conveyance control unit **611** in ACT**23**, and the conveying of the RFID paper P.sub.rf which was stopped at the registration roller **23** is resumed.

(87) In ACT**25**, the processor **61** operates as the writing success/failure determination unit **613** and notifies a writing error (tag writing failure). After that, the processor **61** proceeds to the processing operation of ACT**31**.

(88) The processor **61** starts the conveyance control process illustrated in FIG. **7** in response to an

error notification. First, in ACT41, the processor 61 operates as the RFID paper conveyance control unit 611 and determines whether a sheet with a tag writing error (a writing error paper) reaches a specified position. The specified position can be, for example, a position in front of the flapper 22E or the position of the fixer 28. Determination of whether the specified position has been reached can be performed by using a detector such as a mechanical switch or an optical sensor, or measuring the elapsed time from when the conveying was resumed in ACT23 by using the RTC 70 and calculating the position of the blank RFID paper P.sub.rf based on the elapsed time and the known conveyance speed. If the blank RFID paper P.sub.rf (the writing error paper) has not reached the specified position (NO in ACT 41), the processor 61 repeats the processing operation of ACT 41. In this manner, the processor 61 waits for the blank RFID paper P.sub.rf to reach the specified position.

(89) When the blank RFID paper P.sub.rf reaches the specified position (YES in ACT41), the processor 61 sets the second conveyance path as the conveyance path for the RFID paper P.sub.rf in ACT42. Specifically, the processor 61 switches the flapper 22E in the conveying unit 22 to the upper stage paper discharge tray 29B via the printer interface 66.

(90) In ACT43, the processor 61 determines whether the blank RFID paper P.sub.rf has been conveyed through the second conveyance path and discharged to the upper stage paper discharge tray 29B. If the blank RFID paper P.sub.rf is not yet discharged to the upper stage paper discharge tray 29B (NO in ACT43), the processor 61 repeats the processing operation of this ACT43. In this manner, the processor 61 waits for the blank RFID paper P.sub.rf to be discharged to the upper stage paper discharge tray 29B.

(91) Once the blank RFID paper P.sub.rf is discharged to the upper stage paper discharge tray 29B (YES in ACT43), the processor 61 sets the first conveyance path as the conveyance path for the next RFID paper P.sub.rf in ACT44. Specifically, the processor 61 switches the flapper 22E in the conveying unit 22 of the printer 2 to the lower stage paper discharge tray 29A via the printer interface 66.

(92) In this manner, the blank RFID paper P.sub.rf for which writing of data to the RFID tag failed is conveyed to the upper stage paper discharge tray 29B and, at the same time, the processing operation for the next RFID paper P.sub.rf can be started.

(93) As described above, according to the image forming apparatus 100 according to the second embodiment, since the processing operation for the next RFID paper P.sub.rf is started without waiting for the RFID paper P.sub.rf to be discharged, continuous writing data and printing can be performed at higher speed.

(94) Although the first and second embodiments have been described above, the various embodiments are not limited thereto.

(95) For example, the RFID paper P.sub.rf on which tag writing has been successfully performed and on which image printing has been performed is discharged to the lower stage paper discharge tray 29A and the blank RFID paper P.sub.rf on which tag writing failed is discharged to the upper stage paper discharge tray 29B, but the reverse arrangement may be adopted. In some examples, when the user designates the discharge destination for the RFID paper P.sub.rf, the second paper discharge unit can be set to be any un-designated paper discharge tray by default or by user choice. When there are three or more paper discharge trays, any paper discharge tray can be used as the second paper discharge unit.

(96) Furthermore, in an embodiment, the conveying of the RFID paper P.sub.rf stopped by the registration roller 23 is resumed after waiting for a determination of the success or failure of writing data to the RFID tag is complete, but when the RFID reader/writer 5 is out of order, the success/failure determination cannot be performed, so that the RFID paper P.sub.rf will remain stopped. Therefore, when the RFID paper P.sub.rf remains stopped at the registration roller 23 for more than a certain period of time, the user may be notified of a failure of the RFID reader/writer 5.

(97) In the RFID processing illustrated in FIG. 5, the processing operation writing data in ACT14

and the processing operation acquiring read data in ACT15 may be performed simultaneously with the processing operations in ACT12 and ACT13. In this case, the processing operation of ACT12 is a processing operation that does not determine when the writing position is reached but may be a determination of when a writing completion waiting position has been reached. The writing completion waiting position may be the position of the registration roller 23. In this manner, the conveying of the RFID paper P.sub.rf and the writing data to the RFID tag may be performed simultaneously. Furthermore, the processing operation setting the first conveyance path in ACT17 may be performed after the processing operation for forming the image in ACT18 or the processing operation restarting the conveying of the RFID paper P.sub.rf in ACT19. In this manner, some processing operations may be performed simultaneously, the order of some processing operations may be changed, and the processing contents of some processing operations may be modified.

(98) In an embodiment, a multifunction peripheral (MFP) is used as the example of an image forming apparatus 100, but a printer having no scanner may be adopted instead.

(99) In an embodiment, a program is stored in advance in the ROM 62 or the auxiliary storage device 64 of the control unit 6 of the image forming apparatus 100. In other examples, programs may be transferred separately in a writable storage device and later installed in the image forming apparatus 100 by a system administrator or the like. Programs and the like can be transferred in a removable computer-readable storage medium or by via a network. The computer-readable storage medium may be of any format, such as a CD-ROM or a memory card, that can store a program.

(100) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

## Claims

1. An image forming apparatus, comprising: a paper storage unit; a printing unit configured to print images on a sheet of paper from the paper storage unit; a conveying unit configured to convey sheets one by one along a path from the paper storage unit to the printing unit; a tag reader/writer unit configured to communicate with a wireless tag on a sheet being conveyed by the conveying unit when the sheet is at a communication position on the path before the printing unit; and a control unit configured to: determine whether a writing of first data to a wireless tag on a first sheet has succeeded by controlling the tag reader/writer to read data from the wireless tag on the first sheet, control the printing unit to print an image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet succeeded, control the printing unit not to print any image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet failed, and after it is determined the writing of the first data to the wireless tag on the first sheet failed, execute a retry process by controlling the conveying unit to convey a second sheet from paper storage unit to the communication position on the path and the tag reader/writer to write the first data to a wireless tag on the second sheet.
2. The image forming apparatus according to claim 1, further comprising: a first discharge location for sheets; and a second discharge location for sheets, wherein the conveying unit includes a path switching unit after the printing unit for switching a discharge destination of sheets between the first and second discharge locations, and the control unit is configured to control the path switching unit so sheets on which an image has been printed by the printing unit are discharged to the first discharge location and sheets on which an image has not been printed by the printing unit are discharged the second discharge location.

3. The image forming apparatus according to claim 2, further comprising: a notification unit configured to indicate to a user that a sheet for which the writing of data to the wireless tag has failed has been discharged to the second discharge location.
4. The image forming apparatus according to claim 3, wherein the notification unit comprises a display screen.
5. The image forming apparatus according to claim 2, wherein the first discharge location is a first paper tray, and the second discharge location is a second paper tray.
6. The image forming apparatus according to claim 1, further comprising: a registration roller on the path before the printing unit, wherein, the communication position is at the registration roller.
7. The image forming apparatus according to claim 6, wherein the control unit is configured to control the tag reader/writer unit so as to perform the writing of data to wireless tags on sheets that are temporarily stopped at the registration roller.
8. The image forming apparatus according to claim 1, wherein the wireless tag is an RFID tag.
9. A multifunction peripheral device, comprising: a paper storage unit; a printing unit configured to print images on a sheet of paper from the paper storage unit; a tag reader/writer unit configured to communicate with a wireless tag on a sheet at a communication position on a conveyance path before the printing unit; a first discharge location for sheets; a second discharge location for sheets; a conveying unit configured to convey sheets one by one from the paper storage unit to the printing unit and then to the first or second discharge location; and a control unit configured to: control the tag reader/writer unit attempt a writing of first data to a wireless tag on a first sheet from the paper storage unit at the communication position, determine whether the writing of first data to a wireless tag on the first sheet has succeeded by controlling the tag reader/writer to read data from the wireless tag on the first sheet, control the printing unit to print an image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet succeeded and the conveying unit to convey the first sheet to the first discharge location, and control the printing unit not to print any image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet failed and the conveying unit to convey the first sheet to second discharge location.
10. The multifunction peripheral device according to claim 9, wherein the control unit is further configured to: execute a retry process after it is determined the writing of the first data to the wireless tag on the first sheet failed by controlling the conveying unit to convey a second sheet from paper storage unit to the communication position on the path and the tag reader/writer to write the first data to a wireless tag on the second sheet.
11. The multifunction peripheral device according to claim 9, wherein the conveying unit includes a path switching unit after the printing unit for switching a discharge destination of sheets between the first and second discharge locations.
12. The multifunction peripheral device according to claim 9, further comprising: a notification unit configured to indicate to a user that a sheet for which the writing of data to the wireless tag has failed has been discharged to the second discharge location.
13. The multifunction peripheral device according to claim 12, wherein the notification unit comprises a display screen.
14. The multifunction peripheral device according to claim 9, wherein the first discharge location is a first paper tray, and the second discharge location is a second paper tray above the first discharge location.
15. The multifunction peripheral device according to claim 9, further comprising: a registration roller on the path before the printing unit, wherein, the communication position is at the registration roller.
16. The multifunction peripheral device according to claim 9, wherein the wireless tag is an RFID tag.
17. The multifunction peripheral device according to claim 9, wherein the control unit is further

configured to: after it is determined the writing of the first data to the wireless tag on the first sheet succeeded, control the tag reader/writer unit attempt a writing of second data to a wireless tag on a second sheet from the paper storage unit at the communication position.

18. An image forming apparatus, comprising: a paper storage unit; a printing unit configured to print images on a sheet of paper from the paper storage unit; a conveying unit configured to convey sheets one by one along a path from the paper storage unit to the printing unit; a tag reader/writer unit configured to communicate with a wireless tag on a sheet being conveyed by the conveying unit when the sheet is at a communication position on the path before the printing unit; and a control unit configured to: attempt a writing of first data to a wireless tag on a first sheet at the communication position, determine whether the writing of first data to the wireless tag on the first sheet has succeeded by controlling the tag reader/writer to read data from the wireless tag on the first sheet, control the printing unit to print an image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet succeeded, control the printing unit not to print any image on the first sheet when it is determined the writing of the first data to the wireless tag on the first sheet failed, and after it is determined the writing of the first data to the wireless tag on the first sheet failed, execute a retry process by controlling the conveying unit to convey a second sheet from paper storage unit to the communication position on the path and the tag reader/writer to write the first data to a wireless tag on the second sheet.

19. The image forming apparatus according to claim 18, further comprising: a first discharge location for sheets; and a second discharge location for sheets, wherein the conveying unit includes a path switching unit after the printing unit for switching a discharge destination of sheets between the first and second discharge locations, and the control unit is configured to control the path switching unit so sheets on which an image has been printed by the printing unit are discharged to the first discharge location and sheets on which an image has not been printed by the printing unit are discharged the second discharge location.

20. The image forming apparatus according to claim 19, further comprising: a notification unit configured to indicate to a user that a sheet for which the writing of data to the wireless tag has failed has been discharged to the second discharge location.

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